

TMR sensing as an end-to-end solution

TMR sensing is evolving beyond being just a sensor. It is becoming a full end-to-end solution. A 1000x performance gain is achieved by combining three elements:

- 1. Improved TMR stacks.** Optimised at the sensor level for higher sensitivity
- 2. ASIC integration.** Placed close to the sensor to amplify, filter, and digitise signals at the source, reducing noise
- 3. Artificial intelligence.** Used downstream to distinguish signals of interest from interference, enabling real-time noise reduction

This pipeline approach allows TMR sensors to perform tasks previously impossible outside controlled environments. For example, when worn on the chest, the integrated system can detect arrhythmias or predict cardiac events at home. Other potential applications include gesture recognition, fall prediction in elderly individuals, and monitoring disease progression, all of which can be leveraged through the rapid progress of AI.

once we use them as true analogue sensors that can linearly measure magnetic fields, the game changes. They can pick up biomagnetic signals from the body, such as heartbeats or muscle activity. Early experiments have already shown TMR sensors detecting cardiac signals. The challenge? These tests were usually carried out in laboratories with large, complex setups. Now, the real task is making them small and scalable so they fit into something you and I could actually wear.

This is where progress gets exciting. Today's TMR sensors measure just a few millimetres across, small enough to fit into flexible wearables or even be woven directly into clothing. Unlike traditional sensors, they do not need to touch your skin, opening up new ways to monitor health continuously and comfortably.

Of course, TMR sensors are not here to replace every sensor we use today. They are here to complement them. For example, they will not measure body temperature, but they could track heart rate or even estimate blood pressure simply by sensing magnetic fields from cardiovascular activity. This makes them ideal for next-generation wearables where multiple sensors work together to provide meaningful health insights.

Looking further ahead, another field where TMR sensors could shine is quantum technology. Measuring currents at incredibly small scales requires sensors that are both highly sensitive and compatible with semiconductor processes. This is exactly where sub-10-picoTesla TMR sensors come in. They can detect magnetic fields that are so weak, they are generated by neural or

On-board charger for Electrical Vehicles equipped with Diotec

Practical Thinking: On-board charger for Electrical Vehicles equipped with Diotec three phase bridge rectifier DBI25-16A

An onboard charger (OBC) is a device that is integrated into the electric vehicle (EV) and is responsible for charging the battery pack. It typically operates on AC power and converts it to DC power to charge the battery. The voltage and power range of an on-board charger can vary depending on the specific make and model of the EV. However, most on-board chargers for EVs operate on a voltage range of 100-240 V AC and have a power output range of 3 to 10 kW. Some high-end EVs may have on-board chargers with a higher power output, up to 20 kW or more.



It is important to note that the charging time and capacity of the battery depend on the charger's power output and the battery's capacity. A higher power output charger will generally charge the battery faster, but it may also require a control of the charging process, which can be helpful for managing charging times and optimizing battery life.

Converter applications continuously demand components that offer both high performance and reliability, particularly to withstand increased thermal stresses. To address this, robust input rectification is required capable of handling thousands of volts. The Diotec DBI25-16A three phase in-line bridge rectifier is designed to fulfill the complete need for on-board charger for electrical vehicles. It comes with excellent power output of 40 kW, 25A continuous forward current at 115°C with voltage drop <1.05V and 1600V peak reverse voltage at 5uA. The bridges can be operated at 150°C and protect the circuit against the input surge to till 390A. The case material is UL approved with UL 94V-0 with file E175067.

FOR MORE INFO



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